A Universal Formula for Gravitational Effects Across All Scales Derived from Emergent Gravity in KiCA

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Abstract

The absolute fusion of space, time, and metric within the continuous "Spacetime" manifold is the bedrock of General Relativity, yet it inevitably leads to mathematical singularities under extreme physical conditions. In this paper, we propose a fundamental resolution to the singularity paradox based on the Kizuna-Causality Automaton (KiCA), a pre-geometric, coordinate-free discrete dynamical network. By completely decoupling time, spatial topology, and metric distance, we demonstrate that gravitational effects can spontaneously emerge from the network's underlying topological rewiring without any *a priori* gravitational assumptions. We rigorously define "timed-space curvature" as the normalized graph Laplacian of the local time flow field, proving its equivalence to the Ricci scalar in the weak-field limit. Furthermore, we provide a quantitative analysis of the emergent discrete gravitational dynamics across all scales: at the macroscopic scale, it perfectly reduces to the classical inverse-square law; at the microscopic limit, the interaction naturally saturates and vanishes due to the upper bound of local network dimensionality, thoroughly eliminating the singularity; at the mesoscopic scale, it yields a theoretically falsifiable $1/r^4$ correction term that aligns precisely with the Bardeen regular black hole model. This framework offers a novel, singularity-free mathematical bridge between discrete quantum information and continuous spacetime geometry.

1 Introduction: The Limits of Spacetime Continuum and the KiCA Framework

The absolute fusion of space, time, and metric within the concept of "Spacetime" has served as the bedrock of modern cosmology. Historically, unifying space and time into a single continuum was a stroke of genius; it elegantly resolved the paradoxes of fractured, independent coordinate systems in classical mechanics, providing a cohesive framework for relativistic phenomena.

However, every profound paradigm has its boundary. The *excessive* fusion of these fundamental properties—locking time, spatial topology, and metric distance into a single, rigid tensor—becomes a mathematical straitjacket under extreme physical conditions. It is precisely this over-entanglement that inevitably leads to the mathematical breakdown at singularities. When the system is deprived of the freedom to decouple its dimensions, infinite curvature becomes an unavoidable artifact of the mathematical formulation, rather than a physical reality.

Over the decades, immense theoretical efforts have been dedicated to resolving these singularities, largely by modifying the gravitational action while remaining within the paradigm of continuous pseudo-Riemannian manifolds. For instance, modified gravity theories such as $f(R)$ gravity Starobinsky (1980) and higher-derivative gravity attempt to soften short-distance divergences by introducing higher-order curvature terms. In the context of black holes, phenomenological models like the Bardeen regular black hole Bardeen (1968); Hayward (2006) replace the central singularity with a de Sitter core, effectively modifying the gravitational potential at microscopic scales, though often at the cost of introducing exotic matter that violates strong energy conditions. Meanwhile, major quantum gravity candidates like String Theory Polchinski (1998) and Loop Quantum Gravity (LQG)

Rovelli (2004) introduce extended objects or quantized geometric spectra to cure UV divergences. Yet, they still grapple with the immense complexity of higher-dimensional continuous target spaces or the difficult transition to a smooth macroscopic classical limit. Despite these profound advancements, the fundamental tension between a continuous metric and microscopic discreteness remains unresolved.

To fundamentally break free from this mathematical straitjacket, recent studies Chen (2026) have proposed a discrete dynamical system known as the Kizuna-Causality Automaton (KiCA). Operating without coordinate selections or scale constraints, this framework derives its dynamical laws through underlying graph theory and algebraic mappings. Previous research has shown that this purely relational network, when processing causal flows and local topological rewirings (i.e., causal frustration), naturally gives rise to mathematical formalisms highly consistent with microscopic quantum mechanics. However, that literature primarily focused on micro-scale topological analysis, leaving macroscopic physical phenomena—particularly gravity and relativistic effects—unaddressed.

Building upon this foundation, this paper deeply investigates the intrinsic properties of the KiCA system, demonstrating that gravitational effects can spontaneously emerge from this discrete network without the need for any *a priori* gravitational assumptions. In the vicinity of massive objects, this emergent gravity naturally derives complete relativistic phenomena, including time dilation. Furthermore, we systematically explore the emergent gravitational dynamics across all scales within this system. We show that these dynamics perfectly reduce to classical gravitational formulas at macroscopic scales, while at microscopic scales, they precisely align with the modified equations of the Bardeen model, thereby naturally avoiding short-distance gravitational divergences and thoroughly eliminating the singularity.

2 DEFINITION AND TERMINOLOGY OF KICA

In the following, we adopt the core definitions and notations of the original framework Chen (2026) : **State Space**, **Evolution Rules**, and **Initial Conditions**.

2.1 STATE SPACE & TOPOLOGICAL SKELETON

At any causal slice (i.e., the "present" of the system), the global complete state of the KiCA automaton is strictly defined by a 4-tuple $\mathcal{K}$:

$$\mathcal{K} = (V, E, \Phi, \tau)$$

- V **(Vertices)**: Eternally existing discrete computational units. The vertex set is strictly partitioned into V_A and V_B (with $V_A \cap V_B = \emptyset$). There is a slight asymmetry in their numbers: $|V_A|/|V_B| = 1 + \epsilon$.
- E **(Kizuna Edges)**: Undirected connections between vertices, which are only allowed to be established between V_A and V_B (strict bipartite graph property).
  - *Note: If we strip away the state values and beats, we can define the system's **global topological skeleton** as $G = (V, E)$. For any local vertex subset $U \subset V$, its **closed sub-skeleton** is $G[U] = (U, E_U)$, while an **open sub-skeleton** containing connections to the external environment is denoted as $G_{open}[U] = (U, E_U, \partial U)$, where ∂U is the set of boundary Kizuna edges.*
- Φ **(Causal Potential Function)**: A mapping function $\Phi : V \to \mathbb{Z}$, representing the causal potential currently held by each node.
- τ **(Global Beat)**: The binary clock indicator of the system, $\tau \in \{0, 1\}$.

2.2 EVOLUTION RULES

The operation of the KiCA automaton obeys, and only obeys, the following two underlying algebraic mapping rules. The evolution of the system manifests as a discrete transition from the current state $\mathcal{K}$ to the next causal state $\mathcal{K}'$ ($\mathcal{K} \mapsto \mathcal{K}'$):

- **Rule I: Binary Beat & Causal Drive**
  The direction of causal transfer is determined by τ, and the occurrence of the transfer is driven by the gradient of Φ. We define the causal potential transfer operator $\hat{T}$, which maps the current state value Φ to an intermediate state $\tilde{\Phi}$:
  - **When** $\tau = 0$: The causal direction is $V_A \to V_B$. For every edge (v_A, v_B) in the network, if $\Phi(v_A) \geq \Phi(v_B)$, then v_A transfers 1 unit of Φ to v_B.
  - **When** $\tau = 1$: The causal direction is $V_B \to V_A$. For every edge (v_A, v_B) in the network, if $\Phi(v_B) \geq \Phi(v_A)$, then v_B transfers 1 unit of Φ to v_A.
  - All edges satisfying the conditions complete the transfer **simultaneously** within the current causal step, generating the intermediate state distribution $\tilde{\Phi}$.
- **Rule II: Causal Frustration & Local Rewiring**
  If, after the synchronous update of Rule I, the magnitude relationship of the state values at both ends of a certain edge undergoes a substantial reversal (i.e., before transfer $\Phi(v_i) \geq \Phi(v_j)$, but the intermediate state after transfer is $\tilde{\Phi}(v_i) < \tilde{\Phi}(v_j)$), then this edge is judged to have experienced "causal frustration".
  We define the topological rewiring operator $\hat{R}$ acting on the set $F \subset E$ composed of all frustrated edges: frustrated Kizuna edges must immediately break and forcibly undergo cross-rewiring with another adjacent frustrated edge. The old edges (A_1, B_1) and (A_2, B_2) break instantly and rewire into new edges (A_1, B_2) and (A_2, B_1). After rewiring is complete, a new Kizuna edge set E' is generated.
- **State Update & Beat Flip**
  After completing the above topological rewiring, the global beat automatically flips: $\tau' = 1 - \tau$.
  The system officially updates to a completely new causal state:
  $$\mathcal{K}' = (V, E', \tilde{\Phi}, \tau')$$
  (At this point, $\tilde{\Phi}$ becomes the Φ' of the new state, and the system is ready for the next evolutionary mapping).

2.3 Initial Conditions

The initiation of the automaton only requires assigning a set of initial states $\mathcal{K}_{init}$ that conform to the state space definition. In this minimalist framework, the only global resource that is strictly prescribed and allocated is the total number of initial Kizuna edges:

$$|E_{init}| = \max(|V_A|, |V_B|)$$

Other than this, any microscopic assignment satisfying the basic definitions is legal. The initial topological connection E_{init} can be a randomly matched or highly clustered subgraph; the specific distribution of the initial state values Φ_{init} and the choice of the initial beat τ_{init} are not subject to mandatory restrictions.

3 KiCA Evolutionary Closed-Loop Based on Non-linear Graph Laplacian

We use the **Causal Step Index** $n \in \mathbb{N}$ to denote the discrete evolutionary generations the system undergoes. The complete state of the system is represented by the 4-tuple $\mathcal{K}_n = (V, E_n, \Phi_n, \tau_n)$.

Below, we utilize the language of the Graph Laplacian Chung (1997) to reorganize and strictly formulate the evolutionary closed-loop of KiCA.

3.1 Core Symbols, Functions, and Operator Definitions

To describe the system using algebraic graph theory, we first define the operators and matrices acting on the n-th causal step:

- **Oriented Incidence Matrix B_n:**
The matrix dimension is $|V| \times |E_n|$. Its orientation is entirely determined by the current global beat τ_n. For any Kizuna edge $e = (u, v)$ in the current skeleton E_n:
  - When $\tau_n = 0$, the direction is defined as $V_A \to V_B$.
  - When $\tau_n = 1$, the direction is defined as $V_B \to V_A$.

  If the direction of edge e is $u \to v$, the matrix elements are defined as:
$$(B_n)_{i,e} = \begin{cases} 1, & \text{if } i = u \text{ (potential outflow end)} \\ -1, & \text{if } i = v \text{ (potential inflow end)} \\ 0, & \text{otherwise} \end{cases}$$
- **Discrete Gradient $\nabla_n \Phi_n$:**
Defined as $\nabla_n \Phi_n = B_n^T \Phi_n$. This is a column vector of length $|E_n|$, where its e-th component precisely represents the causal potential difference between the starting and ending points of edge e.
- **Quantized Flux Function $\mathcal{H}(x)$:**
To capture the non-linear characteristic of the rule "if the potential difference is greater than or equal to 0, transfer 1 unit," we define the step activation function:
$$\mathcal{H}(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases}$$
When $\mathcal{H}$ acts on a vector, it is applied element-wise. Therefore, the actual transfer flux vector in the current network is $J_n = \mathcal{H}(B_n^T \Phi_n)$.
- **Non-linear Graph Laplacian $\mathcal{L}_n$:**
Combining the above definitions, we define the causal potential transfer operator of KiCA as:
$$\mathcal{L}_n(\Phi_n) = B_n \mathcal{H}(B_n^T \Phi_n)$$
- **Topological Rewiring Operator $\hat{R}$:**
Acts on the set of edges experiencing causal frustration, executes cross-rewiring, and outputs the new edge set E_{n+1}.

3.2 The Closed-Loop Evolution

The complete process of KiCA transitioning from the causal state $\mathcal{K}_n$ to $\mathcal{K}_{n+1}$ can be strictly formulated into the following four sequential algebraic steps:

Step 1: State Flow
The causal potential undergoes a one-step non-linear Laplacian diffusion on the current topological skeleton, generating the intermediate state $\tilde{\Phi}_n$:

$$\tilde{\Phi}_n = \Phi_n - \mathcal{L}_n(\Phi_n) = \Phi_n - B_n \mathcal{H}(B_n^T \Phi_n)$$

Step 2: Frustration Detection
Calculate the gradient vectors before and after the transfer. Define the frustrated edge set $F_n \subseteq E_n$: For any edge e, if the following conditions are met, then $e \in F_n$:

$$(B_n^T \Phi_n)_e \geq 0 \quad \text{and} \quad (B_n^T \tilde{\Phi}_n)_e < 0$$

(i.e., before the transfer, the potential at the starting point is greater than or equal to the ending point; after the transfer, the potential at the starting point becomes less than the ending point).

Step 3: Topological Rewiring
If $F_n \neq \emptyset$, the rewiring operator $\hat{R}$ intervenes, performing cross-rewiring on the frustrated edges to generate the topological skeleton for the next causal step:

$$E_{n+1} = \hat{R}(E_n, F_n)$$

(If $F_n = \emptyset$, then $E_{n+1} = E_n$).

Step 4: Beat Flip & State Finalization
The global beat flips, the intermediate state solidifies into the new state, and the system completes one causal transition:

$$\tau_{n+1} = 1 - \tau_n$$

$$\Phi_{n+1} = \tilde{\Phi}_n$$

At this point, the system is ready to face the $(n + 1)$-th evolutionary step as $\mathcal{K}_{n+1} = (V, E_{n+1}, \Phi_{n+1}, \tau_{n+1})$.

3.3 Non-linear Characteristics

Through the Laplacian closed-loop formulation described above, KiCA exhibits a distinct characteristic: **Non-linear 1-Laplacian Flow** Hein et al. (2005); Elmoataz et al. (2008).

The classical graph Laplacian describes heat conduction, which smooths out all differences and rapidly drives the system toward heat death. In contrast, KiCA employs a non-linear operator equipped with a step function $\mathcal{H}$. Mathematically, this is analogous to **Total Variation Flow** or the **1-Laplacian operator** on graphs. Its physical significance lies in the fact that the transfer of energy is **quantized** and **threshold-dependent**. This non-linearity not only prevents the system from rapidly smoothing out, but instead maintains sharp local gradients during evolution. This continuously generates frustration, thereby sustaining the perpetual "boiling" and evolution of the system's topological structure.

4 Emergence of "GR-like" Effects in KiCA

In physics, a global "absolute time" does not exist. Time is essentially a **sequence of events** Sorkin (2003); Misner et al. (1973). If no events occur in a local region (i.e., the state remains unchanged and the connections remain unchanged), then for that specific local region, time is "stationary".

We can strictly mathematize this "experiential causal step sequence" and explore the astonishing physical inferences it brings forth.

4.1 Mathematical Definition: Local Proper Time

We treat the global causal step index n as a kind of "Coordinate Time" (which simply serves the convenience of ordering the evolution of the entire system from a God's-eye view).

Now, for each vertex $i \in V$, we define its exclusive **local proper time** (i.e., the experiential causal step sequence), denoted as $s_i(n)$. Initially, the proper times of all vertices are synchronized: $s_i(0) = 0$.

When the global step index transitions from n to $n + 1$, we need to detect whether vertex i has experienced an "event". We define the local neighborhood (connectivity relationship) of vertex i at the n-th step as $\mathcal{N}_i(n) = \{j \mid (i, j) \in E_n \text{ or } (j, i) \in E_n\}$.

We define an **Event Indicator function** $I_i(n)$:

$$I_i(n) = \begin{cases} 1, & \text{if } \Phi_{n+1}(i) \neq \Phi_n(i) \quad \text{(state changes)} \\ & \text{or } \mathcal{N}_i(n+1) \neq \mathcal{N}_i(n) \quad \text{(local topology changes)} \\ 0, & \text{if both state and local topology remain unchanged} \end{cases}$$

Thus, the **local proper time evolution equation** for vertex i is:

$$s_i(n + 1) = s_i(n) + I_i(n)$$

4.2 Physical Intuition: "GR-like" Effects in the Network

Upon introducing this definition, the KiCA system immediately and automatically exhibits fascinating effects analogous to General Relativity (GR):

- **Relativity of Time and Time Dilation**: During the system's evolution, certain regions may possess large causal potential gradients, leading to frequent energy transfers and topological rewiring ("boiling" regions); meanwhile, other regions may have already stabilized in potential with a solidified topology ("cooling" regions). Under the same global step count n, for a vertex in the boiling region, $s_i(n) \approx n$, meaning time flows extremely fast. Conversely, for a vertex in the cooling region, $s_j(n) \ll n$, meaning time flows extremely slowly. **This naturally gives rise to a time dilation effect induced by energy density / gravitational fields!** The more intense the energy exchange in a location, the faster its local time passes.
- **Background Independence**: Leibniz and Mach once proposed that without the motion of matter, time loses its meaning, a philosophy deeply embedded in modern background-independent quantum gravity Rovelli (2004).

  If the entire network falls into dead silence (no potential differences, no frustrations), then for all nodes, $I_i(n) \equiv 0$, and the local proper time s_i of all nodes will permanently stagnate. At this point, even though the global n continues to tick away in the God's-eye view, the internal time of the system has truly "stopped".
- **Event Horizons and Black Holes**: If a subset of nodes in the network becomes completely disconnected from the main network due to topological rewiring, and their internal potentials are equalized, this subgraph will no longer undergo any state or topological updates. Their local proper time will freeze completely. In a causal network, this is equivalent to the formation of an **event horizon**—the evolution of the outside world can no longer affect them, and their time remains eternally stationary relative to the outside world.

4.3 Crucial Caveat: The Decoupling of Time and Spatial Distance

- **Emphasis:** It is important to note that the "GR-like" effects emerging here are completely decoupled from traditional "metric-distance". In the pre-geometric substrate of KiCA, we first discuss topology and time; rigorous curvature calculations will be expanded upon in the next chapter.

5 Timed-Space Curvature in KiCA

5.1 The Straitjacket of GR

In General Relativity (GR), the metric tensor $g_{\mu\nu}$ acts as a dictator. It simultaneously monopolizes the flow of time, the spatial metric-distance, and spacetime curvature. Because these three aspects are forcibly bound by the same mathematical object (the metric) on a fixed, a priori 4-dimensional manifold (x, y, z, t), when matter undergoes extreme collapse and the metric-distance $ds \to 0$, the second derivatives of the metric (which represent curvature) have no other mathematical space to release their energy except by tending to infinity. A singularity is, in essence, the mathematical breakdown of a fixed-dimensional manifold under extreme conditions.

5.2 The Magic of Decoupling

The elimination of singularities in KiCA is not mere semantic wordplay; rather, it introduces genuine physical degrees of freedom that are overlooked by GR.

The underlying substrate of KiCA is a **pre-geometric discrete graph**. Here, **time (update ticks)**, **metric-distance (topological hops)**, and **Timed-Space Curvature** (the graph Laplacian of the time flow field) are **thoroughly decoupled** in their fundamental mechanisms.

The Magic Revealed: The decoupling gives the system a brand new degree of freedom: **Local Network Dimensionality**.

The concept of a dynamic, scale-dependent spacetime dimensionality is highly resonant with modern approaches to quantum gravity, such as Causal Dynamical Triangulations Ambjørn et al. (2005).

Imagine an immense causal potential (energy) concentrated on a single node. In GR, space is crushed (distance goes to zero, curvature becomes infinite); however, in KiCA, this node will frantically establish new connections with surrounding nodes (generating massive amounts of new edges).

The metric distance (hop count) might still be short, but the local dimensionality skyrockets from 3 dimensions to thousands of dimensions.

5.3 The Mathematical Antidote: Rigorous Definition of Normalized Graph Laplacian of Time Flow as Timed-Space Curvature

To connect local proper time with the time flow field, we define the **time flow field** $\gamma_i(n)$ of node i as the rate of change (or difference) of its local proper time $s_i(n)$ relative to the global causal step n:

$$\gamma_i(n) = \frac{\Delta s_i(n)}{\Delta n} \approx I_i(n)$$

In KiCA, the local timed-space curvature $R_i(n)$ of node i is strictly defined as the Normalized Graph Laplacian of the time flow field γ:

$$R_i(n) = \gamma_i(n) - \frac{1}{|\mathcal{N}_i(n)|} \sum_{j \in \mathcal{N}_i(n)} \gamma_j(n)$$

This formula quantifies the difference between the time flow rate at a specific point and the average flow rate of its local neighborhood. Mathematically, this is exactly the discrete generalization of the continuous Laplacian operator ∇^2 onto a dynamic graph network.

- If $R_i > 0$, the local time flow rate is **higher** than the average of its neighbors (it is a local "peak" or source).
- If $R_i < 0$, the local time flow rate is **lower** than its neighbors (it is a local "valley" or sink).

Since curvature is defined as the Normalized Graph Laplacian, its denominator naturally incorporates the node's degree (i.e., the local dimensionality $|\mathcal{N}_i(n)|$).

Theorem 1 (Boundedness of Discrete Timed-Space Curvature and Singularity Avoidance)**.** *Let $\mathcal{K}_n = (V, E_n, \Phi_n, \tau_n)$ be the state of the KiCA system at causal step n. For any vertex $i \in V$, let $\gamma_i(n) \in [0,1]$ denote the local time flow field, and $k_i(n) = |\mathcal{N}_i(n)|$ denote its local degree (effective dimensionality). The discrete Timed-Space curvature $R_i(n)$, defined as the normalized graph Laplacian of $\gamma_i(n)$, satisfies the strict global bound:*

$$\sup_{i \in V, n \in \mathbb{N}} |R_i(n)| \leq 1$$

Furthermore, in the extreme limit where the local causal potential diverges ($\Phi_n(i) \to \infty$), the local rewiring operator $\hat{R}$ drives the local degree to infinity ($k_i(n) \to \infty$). Despite this divergence in effective dimensionality, the curvature $R_i(n)$ remains strictly bounded within $[-1, 1]$, thereby mathematically precluding the existence of geometric singularities.

Proof. By definition, the event indicator function $I_i(n) \in \{0, 1\}$, which implies the time flow field $\gamma_i(n) \approx I_i(n)$ is bounded such that $0 \leq \gamma_i(n) \leq 1$ for all i and n. The discrete curvature is given by:

$$R_i(n) = \gamma_i(n) - \frac{1}{k_i(n)} \sum_{j \in \mathcal{N}_i(n)} \gamma_j(n)$$

Since the second term is a simple arithmetic mean of the neighbors' time flow fields, it must also lie within the interval $[0, 1]$. The difference between two values in $[0, 1]$ is strictly bounded in $[-1, 1]$. Therefore, $|R_i(n)| \leq 1$ holds unconditionally, independent of the magnitude of $\Phi_n(i)$ or $k_i(n)$. □

5.4 Equivalence with General Relativity in the Weak-Field Limit

Any advanced theory must be compatible with its predecessor in the low-energy/weak-field limit. We demonstrate that in the macroscopic limit—where curvature is low and dimensionality stabilizes at approximately 3D—KiCA's graph Laplacian curvature degenerates to the continuous curvature on an Einstein manifold.

In General Relativity (GR), spacetime geometry is described by the metric tensor $g_{\mu\nu}$. In the weak-field and low-velocity approximation (the Newtonian limit), the time component g_{00} is related to the classical Newtonian gravitational potential Φ_N by:

$$g_{00} \approx -\left(1 + \frac{2\Phi_N}{c^2}\right)$$

The flow rate ratio of local proper time (denoted here as τ_{GR} to distinguish it from KiCA's global tick τ) to global coordinate time t is given by $\frac{d\tau_{GR}}{dt} = \sqrt{-g_{00}} \approx 1 + \frac{\Phi_N}{c^2}$.

In our system, the node's time flow rate γ_i corresponds exactly to $\frac{d\tau_{GR}}{dt}$. Therefore, in the macroscopic continuous limit, the time flow field γ is linearly and positively correlated with the gravitational potential Φ_N: $\gamma \propto \Phi_N$.

According to the Einstein field equations, in the weak-field limit, the spacetime scalar curvature (Ricci Scalar) R is proportional to the matter energy density ρ, and the gravitational potential satisfies Poisson's equation:

$$\nabla^2 \Phi_N = 4\pi G\rho \propto R$$

Substituting $\gamma \propto \Phi_N$ into the above equation, we inevitably deduce:

$$\nabla^2 \gamma \propto R$$

Conclusion: Applying the Laplacian operator to the time flow field is physically entirely equivalent to calculating the scalar curvature of spacetime. Our discrete formula $R_i(n)$ is not an artificial construct, but rather a natural corollary of the Einstein field equations applied to a discrete causal network.

5.5 Emergence of Gravitational Effects

This rigorous equivalence proves that in the KiCA system, **gravity does not need to be introduced as a fundamental force**, a perspective shared by the emergent gravity paradigm Verlinde (2011). This actually indicates that the "Graviton" estimated in the original paper is simply the Local Rewiring itself, as dictated by network evolution rule II. As long as there are local differences in causal updates at the substrate of the network (leading to an uneven distribution of γ_i), non-zero graph Laplacian values (i.e., discrete curvature) will naturally emerge. This curvature guides subsequent causal evolution, perfectly manifesting at the macroscopic scale as the GR principle: "spacetime curvature guides matter motion." This provides a solid mathematical bridge for unifying discrete quantum information with continuous spacetime geometry.

Under these conditions, a **dynamic Matter-Geometry Coupling** naturally arises. The renowned physicist John Wheeler once summarized General Relativity in a single phrase Misner et al. (1973): "*Matter tells spacetime how to curve, spacetime tells matter how to move*." The closed loop of KiCA perfectly reproduces this philosophy within discrete mathematics:

- **Spacetime tells matter how to move**: The topological skeleton B_n determines the Laplacian operator $\mathcal{L}_n$, which in turn dictates how the causal potential Φ (matter/energy) flows (Step 1).
- **Matter tells spacetime how to curve**: Once the flow of causal potential Φ generates a local overload (frustration F_n), it directly triggers the $\hat{R}$ operator, causing the topological skeleton E_{n+1} to break and rewire (i.e., a change in timed-space curvature) (Steps 2 & 3).

In the following section, we will provide a rigorous quantitative analysis of these emergent gravitational effects, demonstrating how this framework effectively overcomes the inherent limitations of classical gravitational theories.

6 Quantitative Analysis of Gravitational Effects in the KiCA Dynamical System

6.1 The Singularity Paradox in Classical Gravity

In traditional gravitational calculations, whether using Newton's law of universal gravitation or the field equations of General Relativity, the scope of application is inherently limited by the assumption of a macroscopic continuous manifold. This a priori manifold assumption inevitably leads to severe physical and mathematical paradoxes when dealing with interactions at extremely short distances.

Consider the limiting process of two point particles (or black holes) approaching each other: During the approach ($r \to 0$), the gravitational effect obeys the inverse-square law $F \propto 1/r^2$. As the spatial distance between them approaches zero, gravity diverges and tends to infinity (∞). However, at the exact moment of contact and fusion ($r = 0$), the two entities merge into a single physical system, and the internal relative distance ceases to exist. At this point, their relative gravitational interaction naturally "disappears" or becomes physically undefined.

Mathematically, this process constitutes an extremely pathological discontinuity (including an infinite discontinuity). It is impossible for a physical observable to diverge to infinity at one moment and experience a "cliff-like drop" without any supporting physical mechanism at the very next moment (the moment of fusion). This paradox profoundly indicates: **Treating gravity as a geometric effect or an action at a distance dependent on an a priori continuous metric space fails at the microscopic scale.**

6.2 Discrete Gravitational Interaction Formula in KiCA

Based on the theoretical experience of resolving infinite singularities discussed previously, we abandon the a priori "spacetime manifold" assumption and return to first principles(similar in spirit to causal sets Bombelli et al. (1987) and emergent gravity **?**). In the underlying discrete dynamical system of KiCA, once the appearances of "spacetime curvature" and "action at a distance" are stripped away, the essential characteristics of gravity can be reduced to the following two points:

1. **Clustering of Information**: Gravity is a spontaneous evolutionary tendency of local complex patterns (matter/energy) in the system to **minimize the topological cost of information transmission**.
2. **Purely Kinematic Effect**: Gravity is not an externally applied "force," but an **evolutionary bias** exhibited by the system's underlying evolution rules (Rule II: Causal Frustration and Local Rewriting) when processing high-density information.

In the KiCA framework, there is no a priori spatial distance r, only the **topological number of steps of causal connection**. Therefore, the essence of gravitational interaction can be strictly formulated as follows: **During the iterative evolution of the system, the "causal response delay" between two high-density information regions (masses) undergoes a spontaneous, accelerated reduction.**

To this end, we construct a purely kinematic function through the following three steps to strictly measure this interaction:

Definition 1 (Timed-Space Distance / (Causal Delay Function))**.** *Let G_n denote the global state graph of the system at a discrete causal step n. Let A and B be two disjoint subgraphs ($A, B \subset G_n$) representing distinct localized macroscopic patterns (i.e., mass regions).*

*The **Causal Delay Function**, denoted by $\rho(A, B, n)$, is defined as the minimum number of global evolution steps required for a localized perturbation (e.g., a discrete state flip) originating within the interior of A at step n to causally propagate and induce a non-trivial state transition at the boundary ∂B of pattern B.*

*Formally, this function quantifies the pure **timed-space distance** between the two subgraphs in the discrete dynamical system. By definition, it maps to the set of strictly positive integers:*

$$\rho(A, B, n) \in \mathbb{N}^+.$$

6.2.1 Kinematic Mass and Equivalence to Rest Mass

Under first principles, gravitational mass is equivalent to the "state update activity" or "information processing density" of a local region. We define the **Kinematic Mass** $M(A)$ of pattern A as the number of nodes undergoing state changes per unit evolution step in that region (i.e., computational complexity).

Meanwhile, recalling the strict definition of the system's **Rest Mass** in the KiCA framework: Given the total number of fundamental oscillatory closed loops inside a subsystem (i.e., the trace of the square of the internal causal operator), the square of the rest mass m^2 of the system is directly defined as:

$$m^2 \propto \mathrm{Tr}(\hat{C}_{\text{int}}^2) \quad \Longrightarrow \quad m \propto \sqrt{\mathrm{Tr}(\hat{C}_{\text{int}}^2)}$$

Here, we state the following theorem (the rigorous proof is deferred to subsequent work):

Theorem 2 (Emergent Mass Equivalence)**.** *Let $A \subset V$ be a localized topological pattern within the KiCA state space. Let $m_A \propto \sqrt{Tr(\hat{C}_{int}^2)}$ denote the rest mass of A, characterized by the density of its internal causal closed loops, and let $M(A)$ denote its kinematic mass, defined as the rate of local state updates per global causal step Δn. Under the non-linear graph Laplacian evolution defined by Rules I and II, the convergence flux of the external causal gradient upon A is strictly isomorphic to its internal closed-loop density. Consequently, the kinematic mass and the rest mass are mathematically equivalent:*

$$M(A) \equiv m_A$$

Remark: Unlike classical mechanics or General Relativity, where the equivalence between inertial (kinematic) and gravitational (rest) mass is postulated as an *a priori* empirical principle requiring experimental verification, Theorem 2 demonstrates that this equivalence is an inevitable mathematical corollary emerging directly from the underlying graph-theoretic evolution rules of the KiCA framework. In the subsequent discourse, we will uniformly utilize the term "mass" to denote this unified quantity.

6.2.2 The Discrete Interaction Function

In a "trivial vacuum" lacking highly active patterns, the evolution of the causal delay ρ between two probe points with respect to the step sequence n is linear (constant velocity, zero acceleration).

To rigorously quantify the emergent gravitational dynamics between localized structures, we introduce the discrete gravitational function. Let the system state at a discrete time step n be represented by a global causal graph G_n. Suppose A and B are two distinct, disjoint subgraphs ($A, B \subset G_n$). Let $\rho(A, B, n)$ denote the timed-space distance between subgraphs A and B at step n. We define the **Discrete Gravitational Function**, denoted by $\Gamma(A, B, n)$, as the negative second-order discrete time derivative of this density:

$$\Gamma(A,B,n) = -\frac{\Delta^2 \rho(A,B,n)}{\Delta n^2} = -\Big[\rho(A,B,n+1) - 2\rho(A,B,n) + \rho(A,B,n-1)\Big]. \qquad (1)$$

Physical Interpretation: In this graph-theoretic formulation, the second-order central difference operator $\Delta^2/\Delta n^2$ captures the discrete effective acceleration between the subgraphs A and B over the causal evolution steps. The negative sign explicitly ensures that a purely attractive interaction—where the relational density between the subgraphs concentrates, or their effective graph distance decreases—yields a positive gravitational function ($\Gamma > 0$).

Physical Intuition:

- If $\Gamma = 0$, it indicates that the causal distance between A and B changes at a constant rate or remains stationary, and the system exhibits no gravitational interaction.
- If $\Gamma > 0$, it indicates that ρ is accelerating its decrease. This means that when processing the high-density information flow between A and B, the underlying automaton **"devours" or "folds" the intermediate vacuum nodes** through topological rewiring (Rule II). This purely kinematic effect of accelerated approach on the causal graph emerges macroscopically as "gravitational attraction."

6.3 Asymptotic Behaviors of the Discrete Gravitational Formula

This section will prove that this discrete gravitational formula is completely consistent with classical Newtonian/relativistic gravity in the macroscopic limit, while naturally resolving the singularity in the microscopic limit.

6.3.1 The Inverse-Square Law and Singularity Resolution in the Macroscopic 3D Limit

Initial Conditions Setup:

1. **Number of Nodes (Mass)**: Patterns A and B contain N_A and N_B active nodes, respectively, with $N_A, N_B \gg 1$ (macroscopic mass).
2. **Spatial Dimension**: Assume the macroscopically emergent topological dimension is 3. In graph theory, this means that starting from any node, the number of surface nodes $S(\rho)$ of the "causal sphere" reachable in ρ steps satisfies the square growth law: $S(\rho) \approx 4\pi\rho^2$.
3. **Current State**: At step n, the timed-space distance between A and B is ρ_n.

Step 1: Calculating Information Flux
Pattern A generates N_A state flips (information packets) within each evolution step $\Delta n = 1$. After ρ_n steps, these information packets are uniformly distributed on a causal sphere of radius ρ_n. At this time, the **local information density (flux)** $\mathcal{D}_A$ received at the location of pattern B from A is:

$$\mathcal{D}_A = \frac{N_A}{S(\rho_n)} = \frac{N_A}{4\pi\rho_n^2}$$

Step 2: Calculating Causal Rewiring Rate
In the underlying rules of KiCA, when an active node receives high-density information, the system has a probability κ (microscopic topological coupling constant) to establish a shortcut or devour a vacuum node. At step n, the **total number of effective rewiring events E_{rewire}** occurring between the N_B nodes of pattern B and the information flow is:

$$E_{\text{rewire}} = \kappa \cdot N_B \cdot \mathcal{D}_A = \frac{\kappa}{4\pi}\frac{N_A N_B}{\rho_n^2}$$

Step 3: Deriving the Macroscopic Gravitational Formula
The contribution of each rewiring event to the shortening of the macroscopic distance is denoted by a minuscule quantity ϵ. The second-order difference (acceleration) of the causal distance driven by rewiring events is:

$$\Gamma(A, B, n) = \epsilon \cdot E_{\text{rewire}} = \left(\frac{\epsilon\kappa}{4\pi}\right)\frac{N_A N_B}{\rho_n^2}$$

If we define the macroscopic constant $G_{\text{KiCA}} = \frac{\epsilon\kappa}{4\pi}$, we immediately obtain:

$$\Gamma(A, B, n) = G_{\text{KiCA}}\frac{N_A N_B}{\rho_n^2}$$

Conclusion 1: Mathematically, this is **perfectly equivalent to Newton's law of universal gravitation**. However, its physical connotation has been completely reconstructed: mass is replaced by information processing activity, absolute distance is replaced by system evolution steps, and gravity itself becomes the second-order difference of the topological contraction of the causal graph.

Step 4: Singularity Resolution at the Microscopic Limit
When A and B are extremely close, the causal distance ρ_n approaches its discrete lower bound $\rho_{\text{min}} = 1$. At this point, the area of the causal sphere no longer obeys the $4\pi\rho^2$ rule of continuous manifolds, but degenerates to the local degree of the graph nodes, denoted as $k = |\mathcal{N}_i(n)|$.

Crucially, as long as a causal connection exists, this local degree is a strictly positive integer ($k \geq 1$). The information flux therefore no longer diverges but reaches a **bounded saturation value**: $\mathcal{D}_{A(\text{max})} = N_A/k$. Since the macroscopic masses N_A and N_B are finite constants, the gravitational interaction is strictly bounded by a finite maximum:

$$\Gamma_{\text{max}} = \epsilon\kappa \cdot N_B \cdot \frac{N_A}{k} \leq \epsilon\kappa N_A N_B$$

Furthermore, when A and B completely merge (ρ stabilizes at 1 and no longer changes, i.e., $\rho_{n+1} = \rho_n = \rho_{n-1} = 1$), substituting into the second-order difference formula yields:

$$\Gamma = -(1 - 2(1) + 1) = 0$$

Conclusion 2: Gravity is no longer inverse-square at extremely small scales, but naturally saturates due to the strictly positive lower bound of the local network degree. When patterns completely merge, the internal relative causal delay disappears, and gravity strictly returns to zero (asymptotic freedom). The singularity is thoroughly eliminated in the discrete network.

6.3.2 GRAVITATIONAL CORRECTIONS AT THE MESOSCOPIC SCALE

At the mesoscopic scale, the "saturation effect" of the discrete network can be continuously approximated by introducing an effective distance $r_{\text{eff}} = \sqrt{r^2 + r_0^2}$ (where r_0 is the underlying minimal fundamental scale corresponding to $\rho_{\text{min}} = 1$, such as the Planck length).

The effective gravitational formula at this time can be written as:

$$F_{\text{KiCA}} = G\frac{m_1 m_2}{r_{\text{eff}}^2} = G\frac{m_1 m_2}{r^2 + r_0^2}$$

Performing a Taylor expansion in the region $r > r_0$:

$$F_{\text{KiCA}} = G\frac{m_1 m_2}{r^2}\left(1 + \frac{r_0^2}{r^2}\right)^{-1} = G\frac{m_1 m_2}{r^2} - G\frac{m_1 m_2 r_0^2}{r^4} + \mathcal{O}\left(\frac{1}{r^6}\right). \tag{2}$$

This expansion clearly separates into two parts:

1. **Dominant term**: $G\frac{m_1 m_2}{r^2}$, perfectly compatible with macroscopic Newtonian gravity.
2. **KiCA correction term**: $-G\frac{m_1 m_2 r_0^2}{r^4}$.

Physical Significance and Verification Prospects:
This correction term is mathematically highly consistent with the Regular Black Hole Metric artificially constructed by Bardeen Bardeen (1968); Ayón-Beato & García (2000) to eliminate black hole singularities. However, in KiCA, it is not an artificially pieced-together phenomenological term, but an inevitable corollary of discrete topological evolution. The negative sign of the correction term indicates that at mesoscopic/microscopic scales, the true gravitational strength will be weaker than predicted by classical theories, exhibiting an anomalous $1/r^4$ decay rate. This provides a clear theoretical prediction for experimental verification of gravitational corrections at micro/nano scales in the future Kapner et al. (2007); Adelberger et al. (2003)..

6.3.3 SUMMARY OF THIS CHAPTER

The discrete gravitational formula under the KiCA framework has the following three core advantages:

1. **No a priori geometry required**: Dynamics are deduced solely by calculating the propagation steps of information in the discrete network (Timed-Space) and node rewiring probabilities.
2. **Natural emergence of the inverse-square law**: In the macroscopic limit ($\rho \gg 1$), based on the square growth law of the network's surface area, the $1/r^2$ gravitational form is naturally derived.
3. **Built-in singularity resolution mechanism**: In the microscopic limit ($\rho \to 1$), constrained by the upper bound of degrees in the discrete network, gravity naturally saturates and ultimately returns to zero (asymptotic freedom), completely bypassing the divide-by-zero catastrophe caused by continuous manifolds.

7 CONCLUSION, LIMITATIONS, AND FUTURE WORK

Conclusion This paper systematically demonstrates how the KiCA framework achieves the emergence of time and timed-space curvature through a purely discrete relational network. By introducing "local network dimension"—a degree of freedom largely overlooked in traditional theories—this model naturally resolves gravitational singularities and derives a discrete gravitational calculation formula applicable across all scales. This formula is not only macroscopically equivalent to classical theories, but also provides falsifiable gravitational correction terms at the mesoscopic scale.

Limitations The current argumentation in this paper primarily focuses on proving the mathematical self-consistency of the underlying model (i.e., demonstrating that singularities leading to mathematical breakdown will not occur within the system). However, at specific astrophysical scales, the significant differences and constructive predictions between this model and General Relativity in the strong-field limit (such as accretion disk dynamics near black hole event horizons, gravitational wave waveforms, etc.) still require further supplementation and numerical simulation verification by the astrophysics community.

Future Work

1. Throughout our derivation, we have revealed a profound physical fact: "Metric-Distance" is fundamentally decoupled from "Timed-Space Curvature" at the underlying mechanistic level. Therefore, this paper has not yet engaged in an in-depth discussion on the generation mechanism of metric distance. Exploring various self-consistent mechanisms for "Metric Emergence" will become the core theme of our next independent study.
2. *Exploration of computational complexity and dynamical capabilities*: During the analysis of the KiCA dynamical system's updating process, we noted its inherent and significant nonlinear characteristics. As an underlying discrete computable model, intuitively, its dynamically reconfigurable network topology endows it with richer dynamical behaviors and greater computational power than traditional cellular automata on fixed grids (such as Conway's Game of Life Berlekamp et al. (1982)). Recent advances in computer science, such as Causal Graph Dynamics Arrighi & Dowek (2012), have shown that systems with time-varying topologies possess unique computational properties. Future research requires more specific quantitative studies from the perspective of computational complexity theory, rigorously proving its Turing completeness, and exploring whether it possesses unique advantages over classical computational models in simulating fundamental physics Wolfram (2002).

8 REMARKS

As a final remark, the reader may observe that the mathematical apparatus employed throughout this paper is remarkably simple, eschewing the highly convoluted tools traditionally required in the pursuit of singularity resolution. However, we must emphasize that this simplicity in derivation is by no means a lack of mathematical rigor.

Rather, it is the direct result of bypassing a profound conceptual blind spot in our choice of problem-solving paradigms. Historically, countless brilliant minds have followed a standard theoretical sequence: first, enter a coordinate-dependent, fused "Spacetime" continuum; next, seek a coordinate-independent metric and introduce complex tensor calculus to eliminate the artifacts of coordinate selection; and finally, calculate the curvature. It is precisely within this rigid, coordinate-laden metric space that the mathematics inevitably breaks down at extreme scales.

We took a fundamentally different path. We realized that what is physically essential is the **"timed-space curvature"** or **"timed-space distance"** itself. By choosing *not* to enter this coordinate-dependent space in the first place, we completely bypass the traditional metric space and its cumbersome tensor machinery. We compute the gravitational effect directly from a decoupled, coordinate-free foundation. By not forcing our calculations through the bottleneck of a fused Spacetime metric, the Gordian knot is naturally untied. The simplicity of the mathematics presented here is not a compromise, but a reflection of the underlying physical reality: when we take the direct route to compute gravitational effect, the mathematics naturally returns to elegance and clarity.

REFERENCES

Eric G Adelberger, Blayne R Heckel, and Ann E Nelson. Tests of the gravitational inverse-square law. *Annual Review of Nuclear and Particle Science*, 53(1):77–121, 2003.

Jan Ambjørn, Jerzy Jurkiewicz, and Renate Loll. Spectral dimension of the universe. *Physical Review Letters*, 95(17):171301, 2005.

Pablo Arrighi and Gilles Dowek. Causal graph dynamics. In *Automata, Languages, and Programming: 39th International Colloquium, ICALP 2012*, pp. 54–66. Springer, 2012.

Eloy Ayón-Beato and Alberto García. The bardeen model as a nonlinear magnetic monopole. *Physics Letters B*, 493(1-2):149–152, 2000.

James M Bardeen. Non-singular general-relativistic gravitational collapse. In *Proceedings of the International Conference GR5, Tbilisi, USSR*, volume 174, 1968.

Elwyn R Berlekamp, John H Conway, and Richard K Guy. *Winning Ways for your Mathematical Plays, Volume 2*. Academic Press, 1982.

Luca Bombelli, Joohan Lee, David Meyer, and Rafael D Sorkin. Space-time as a causal set. *Physical review letters*, 59(5):521, 1987.

Jingwei Chen. Emergence of physics from the kizuna-causality automaton. Zenodo, 2026. doi: https://doi.org/10.5281/zenodo.20156793.

Fan R. K. Chung. *Spectral Graph Theory*, volume 92. American Mathematical Society, Providence, RI, 1997.

Abderrahim Elmoataz, Olivier Lezoray, and Sébastien Bougleux. Nonlocal discrete regularization on weighted graphs: a framework for image and manifold processing. *IEEE Transactions on Image Processing*, 17(7):1047–1060, 2008.

Sean A Hayward. Formation and evaporation of regular black holes. *Physical Review Letters*, 96 (3):031103, 2006.

Matthias Hein, Jean-Yves Audibert, and Ulrike von Luxburg. Graphs, p-laplacians and total variation. In *Proceedings of the 18th International Conference on Neural Information Processing Systems (NIPS)*, pp. 521–528, 2005.

DJ Kapner, TS Cook, EG Adelberger, JH Gundlach, BR Heckel, CD Hoyle, and HE Swanson. Tests of the gravitational inverse-square law below the dark-energy length scale. *Physical Review Letters*, 98(2):021101, 2007.

Charles W Misner, Kip S Thorne, and John Archibald Wheeler. *Gravitation*. W. H. Freeman, San Francisco, 1973.

Joseph Polchinski. *String theory: Volume 1, An introduction to the bosonic string*. Cambridge university press, 1998.

Carlo Rovelli. *Quantum Gravity*. Cambridge University Press, Cambridge, 2004.

Rafael D Sorkin. Causal sets: Discrete gravity. *arXiv preprint gr-qc/0309009*, 2003.

Alexei A Starobinsky. A new type of isotropic cosmological models without singularity. *Physics Letters B*, 91(1):99–102, 1980.

Erik P Verlinde. On the origin of gravity and the laws of newton. *Journal of High Energy Physics*, 2011(4):1–27, 2011.

Stephen Wolfram. *A New Kind of Science*. Wolfram Media, 2002.